

Preface

GLAZING: glaz·ing ['gleɪzɪŋ]. Generally used to refer to installation of windows or application of glaze in baking or pottery. However, in slang, a new definition has surfaced.

Glazing: To over-compliment something— typically with an almost hyperbolic adoration. To have an excessive bias towards others, often to the point of purposely ignoring all negative traits. Tends to come off as fake or ‘cringe’.

And one thing that our team has seen as being ‘glazed’ are several forms of renewable energy.

We’ve all heard that spiel— use solar energy, use wind energy, use this and that energy because fossil fuels are horrible for

the environment and these energies are far better.

But the realities are that each of these so-called ‘better’ energies come with their own, debilitating, problems— because sometimes, something unideal is painted as so perfect that no one stops to think about how it really *isn't*; because the idea of these ones being perfect and these ones being bad is so ingrained in our minds that we don't see any other possible reality— because the nuclear disasters have been so famous that everyone believes that all nuclear plants will result in disaster and *solar energy couldn't possibly cause as much trouble, could it?*

Or because wind turbines seem so beautiful and efficient that *surely* they must be brilliant and wonderful and amazing, right?

But in this booklet and during this exhibition, we will speak a little about the excessive glazing of common renewable energies and the downplaying and obscurity of several alternatives that could work just as well as— if not better than— solar energy or wind energy. Alternatives that would be more efficient, more effective, easier to implement on large scales.

We'll give some information on each energy and its issues as well as each alternative renewable energy and why they would be good alternatives, backed with proof in the form of diagrams and calculations plus real scientific explanations as to why we believe each alternative is better than or could work well alongside the overused and glazed most common ones, as well as some reasons— if applicable— why these alternatives haven't caught on.

We will dissect the duality of various types of energies, their upsides and downsides alike, and explain the importance of different energies being used in tandem for a greener and cleaner future.

All current versions of this booklet are available on our website at:
<https://lunrnightngale.github.io/exhibition2025/>

Solar

Solar energy, the one at the helm of the renewable energy boat. The figurehead, the captain, and the main driver of renewable or clean energy— although that doesn't necessarily mean it's bad, considering it is one of the most promising renewable energy technologies.

Solar energy works through sunlight— rays of sunlight hit solar electric or, more technically, photovoltaic panels made of photovoltaic cells coated, wired, and encapsulated together. When photons in sunlight hit photovoltaic material in cells, which are usually highly purified silicon, electrons are dislodged from the material, generating a charge that is stored in batteries or sent to the power grid.

The word solar is derived from the Latin ‘solaris’ meaning of, pertaining to, or determined by the sun, while photovoltaic is derived from Latin ‘photo’ meaning light and voltaic from the name of Alessandro Volta, who perfected the chemical process used in electric batteries.

The photovoltaic cells that make up solar panels can be divided into three generations.

First Generation: They are made from crystalline silicon and currently the most efficient solar panels available for residential use.

Crystalline silicon-based solar cells are the most common, cost-effective and stable as of now. The oldest and most popular and efficient cells as of now are still monocrystalline silicon cells made from thin wafers of silicon crystals sliced from large

single crystals grown under controlled conditions. They have a relatively high efficiency of up to 24.2%, and despite originally being more expensive, the prices have fallen a great deal. However, their efficiency is lost as the temperature increases above 25C.

The cheaper but less efficient polycrystalline cells, on the other hand, are produced in moulds with multiple silicon crystals that grow on each other to interlock properly. Their efficiency is up to 19.3%.

Meanwhile, the less efficient amorphous silicon cells have thinner silicon layers on a backing substrate of different materials—less efficient, up to 10%, but panels can be made flexible.

The last type is relatively new—hybrid silicon cells projected to be more efficient and less likely to lose efficiency in higher

temperatures that take advantages from both the mono and polycrystalline cells as well as the amorphous silicon cells.

Second Generation: Usually called thin-film solar cells, these usually are made from layers of semiconductors only a few micrometres. There are essentially three types of solar cells in this category.

Amorphous silicon is considered the safest, most stable and cost-effective cell in this category, as well as the most common.

CdTe, or cadmium telluride, is cheaper to produce— but cadmium is toxic.

CIGS, or copper indium gallium selenide, is flexible and high efficiency but expensive.

GaAs, or gallium arsenide, is extremely efficient but is costly and poses arsenic poisoning risks.

Third Generation: Characterized by a broad spectrum of advancements, the third generation of solar cells seeks to overcome the shortcomings and limitation present in the previous generations of technologies. Among these are the following.

Quantum Dot Solar Cells (QDSCs) utilise quantum dots, or nanocrystals of semiconductors, as light-absorbing components that provide the possibility of tunable bandgaps and through that better efficiency.

Perovskite Solar Cells (PSCs), one of the most popular of the third-generation cells, use perovskite for a rapid increase in efficiency up to and over 25%, exceeding previous theoretical limits. Despite the trade-off in durability, PSCs are fast developing.

Organic Photovoltaics (OPV) make use of organic polymers to harvest light to be more environmentally friendly.

Dye-Sensitized Solar Cells (DSSCs) contain a photosensitive dye that helps to absorb sunlight. They are lightweight, flexible, and have potential for inexpensive manufacture, though their efficiencies lag behind.

From these features so far, we can deduce solar energy is overall a solid energy source, but due to its variability, temperature fluctuations and sunlight as well as geographical location can damage effectiveness.

Furthermore, efficiency is an issue—for instance, to power the city of Toronto, more than 4.56 billion panels would be needed, while the land area required would span over 7,700 square kilometres—over 11 times the

size of Toronto's land area. As solar panels are not stackable vertically and must be placed in one layer, based on conditions and energy use it could take a massive number and amount of space to properly power urban areas.

Some other issues that come up are toxicity— for instance, gallium arsenide contains arsenic that is chemically bonded and encapsulated and would normally be safe, but if panels are damaged, arsenic can leach into soil or water and cause poisoning, harming nearby life.

With degradation and slow damage, chemicals can leach out of the solar panels and enter the surroundings, causing negative effects.

However, solar panels' durability and efficiency have been making drastic improvements in the past few decades.

Low-quality panels also only last about five years due to poor encapsulation, UV damage, and weathering, degrading faster and reducing efficiency significantly.

High-quality panels, on the other hand, have better protective coatings and durable materials, with more gradual efficiency loss and average lifespan of 15-25 years on average.

Degradation generally happens due to UV radiation breaking down materials, thermal cycling causing microcracks, or moisture leading to corrosion.

Solar energy is renewable, clean, and scalable for homes and small businesses—but it is inefficient for large-scale urban energy needs, has some toxicity risks from arsenic and cadmium, and has significant unavoidable maintenance and replacement costs over time. It is best used as a

supplementary or distributed source and not as the sole provider for major cities.

Despite this, recent innovation in solar fields proves promising, and solar energy may soon become more prevalent on a global scale, contributing ever the more to our greener and cleaner future, especially with further engineering.

Biomass

Biomass energy is a renewable energy that utilizes organic material, which is converted usually via direct combustion into usable heat and electricity. It may also be bacterially decomposed to produce gas fuels or converted into liquid or gas fuels through gasification. Organic material may be of plant, animal, or other organic sources with unique characteristics and efficiency.

Biomass is an essential fuel in many developing countries, especially for domestic purposes such as cooking or heating.

Harvestable by many means, biomass energy has traditionally been utilized through direct combustion, especially of ethanol and plant or animal biomass like wood or tallow— first used directly as flame to heat or light areas, then later on to produce

concentrated steam that can be used to produce energy through turbines or engines.

More modernly, organic waste can be collected and decomposed by anaerobic bacteria to produce renewable gas, breaking down organic materials into simpler substances that can then be processed further. This takes place in several steps.

First, exoenzymes from bacteria, protozoa, and fungi, as well as water, are used to break down cellulose, starch, and simple sugars into mono-sugars and fatty acids in a process known as hydrolysis.

Then, acidogenesis occurs—soluble monomers are converted into small organic compounds. Acidogenesis intermediates are then attacked by acetogenic bacteria to produce acetic acid, carbon dioxide and hydrogen in a process known as acetogenesis.

Finally, bacteria contribute to methanogenesis, the last phase, to react into methane.

Gasification can also be used, transforming carbonaceous materials into syngas, a combustible gas.

The process occurs with the biomass first dried, then heated to flash point in the absence of oxygen, cumbering combustion—this produces volatile gases and char and is known as pyrolysis.

The char is partially combusted with a limited amount of oxygen to generate heat, and the remaining char reacts with steam and carbon dioxide to produce syngas. However, the composition of biomass used can affect the efficiency and quality of produced syngas.

It goes back the farthest chronologically of all energy sources— humans first used biomass as an energy source over 230,000 years ago. Biomass is also something that predates the human race as a whole— wildfires run on combustion of wood or other biomass, for instance.

Ever since the human race discovered fire through combustion of wood, we have used it for our benefit— for warmth, protection from enemies, illumination and more. The use of biomass energy has paved the way for fossil fuels and other fuel sources harnessed through combustion and has also been innovated and developed over the centuries.

Since the very first usage of biomass as combustible fuel, humans have begun to use biomass in all kinds of ways, not just energy— as liqueur, as oil, and many more

forms. Biomass is used all over the world in various forms, and it truly speaks to its versatility and benefit to us that it is still relevant in the energy scene today.

The earliest records of ethanol date back to around 1100, when it was first isolated from wine to use in combustion with low heat. By the 1300s with the development of cooling coils ethanol could be distilled to nearly pure, then gaining popularity in the lighting and cooking scene— used in oils and stoves for heat. It is also believed that Sumerians and Ancient Egyptians also used animal and vegetable oils, another form of biomass, for purposes ranging from perfumes and spiritual purification to more important cooking, heating, and lighting.

As heating and lighting developed more, especially around the 1800s, not only oils but

also turpentine and other alcohols were used for heat and light.

While raw pine sap was used a lot for ship-making from the 1700s to mid-1900s, it produced many valuable chemicals and compounds, one of which was turpentine.

In 1826, Samuel Moray used ethanol and turpentine to power the prototype of an internal combustion engine, which sparked a long trend of using ethanol fuel to power industrial machinery. Due to the rise of petroleum as an energy source, however, ethanol as a power source slowly grew unpopular.

Later on, though, bioethanol resurged as a fuel source in the 1900s after the First World War prompted scarcity in fossil fuels, petroleum included. High demand for ethanol returned, only to be again eclipsed by coal and crude oil from mining,

commercialization, and growing globalization.

In the 1970s, however, a fuel crisis prompted governments and academia to return to renewable energy sources like solar energy, geothermal energy, et cetera. The widespread exploration of these sources acted as the groundwork for many modern renewable or green energies.

Biomass through combustion was later linked to climate change— combustion causing carbon emissions that scientists are trying to cut back on today, as well as the harvesting of plant biomass from trees contributing to global deforestation.

With innovation, however, we have more ways to harness biomass for energy other than relying on wood sources and combustion. Biomass energy is proving promising to reduce pollution as well.

Biomass energy can be produced from agricultural or non-agricultural waste like animal manure, food waste, human sewage, et cetera. Agricultural crops like corn, soybeans, or kelp also work, as well as wood and related wood processing waste, and municipal solid waste.

However, there are some disadvantages.

Using too much will damage our biosphere, research showing that we should only be using 10% of plant-produced biomass.

Studies from 2023, however, show that we have been using 30% of all annually produced plant biomass— this could lead be environmentally detrimental due to plant biomass being the basis of food chains, overuse of freshwater that is scarce as-is, mass deforestation, et cetera.

Too much consumption can also lead to increased prices for resources and redirection of vital resources used in food or other purposes— bioethanol, for instance, the earliest form of ethanol produced and utilised by humans and chemically indistinguishable from synthetic ethanol, is and was commonly used for transportation, as well as an additive to conventional fuels to reduce greenhouse gas emissions due to unsuitability as a pure energy source, but is criticised for large consumption of food resources.

Direct combustion or harvesting via combustion can also release harmful pollutants like carbon dioxide and other carbon emissions that contribute to global warming.

On the other hand, biomass can be a clean and reliable energy source that can be

continually produced and harvested in a way as harmless environmentally as possible.

With responsible usage in moderation, as well as further innovation into clean ways to convert biomass into energy, it may prove exceedingly helpful to our integration of and eventually complete or near-complete migration to renewable and clean energy sources, contributing to a cleaner Earth.

Wind

Picture this— a turbine. Doesn't matter what kind.

Most likely, you thought of a large, three-bladed white windmill-type turbine—but this is only one type of wind turbine, and wind energy is only one reason to use turbines.

So, what is wind energy?

Wind energy is the production of electricity via harvesting and converting the kinetic energy of wind. Although, like solar energy, this is also a variably available energy source reliant on weather, it is less intermittent as the wind and wind flow is a product of the uneven absorption of heat by the Earth's surface causing air to flow different places based on temperature and

pressure. It is also influenced by the Coriolis effect caused by the rotation of the Earth.

Wind energy and wind turbines work via aerodynamic blades. Wind flows through the blades, causing unequal air pressure on different sides of the blade. This disparity creates lift and drag, but the lift is stronger than the drag due to the design, causing the rotor to spin. The rotor connects to a generator which in turn translates aerodynamic force into electricity.

There are several different types of generators used, but they vary based on types of wind turbine. Here are some examples:

Induction Generator: This type of generator is asynchronous—the waveform generated is asynchronized to the rotational speed. They can function as a motor or generator—in the former case, the rotor

spins slower than the synchronous speed of the field; the latter is similarly vice versa. It uses torque generated by the asynchronous speed to induce voltage. They can be singly or doubly fed.

Permanent Magnet Generators:

Permanent Magnet Generators function by rotating high-strength permanent magnets, causing a magnetic field to pass through copper wire coils in a stationary component called a stator. The change in magnetic flux induces an electromotive force, resulting in an alternating current in the coils.

Now for some types of turbines:

Horizontal Axis: The most overwhelmingly abundant type of wind turbines used globally, they usually have three blades and function upwind, pivoting at the top of the tower.

Vertical Axis: They have omnidirectionally shaped blades, so they can function without being adjusted.

Turbines can be in many places, and here are the main areas:

Land-Based: They range from 100 kilowatts to several megawatts. They are based on land, and often grouped together to form wind plants, providing the electrical grid with bulk power.

Distributed: These are installed on the “customer” side of the electric meter— they are near to or at the location where their energy will be used. They are typically small— below 100 kilowatts— and used for residential, agricultural, and small commercial and industrial location.

Offshore: Self-explanatory, offshore wind turbines are built off the shore of

landmasses, and they tend to be massive—the big components have less transportation difficulties than land-based installations. These can capture strong ocean winds, producing huge amounts of energy.

Wind energy has many advantages. For instance, it can create well-paying jobs—in the United States alone, an estimated 150,000 people are working in the wind industry across all 50 states—and that number continues to grow. Wind turbine service technicians are the fastest growing job in the United States of the decade. Offering a diverse array of career opportunities from blade fabricator to asset manager, the wind industry has the potential to support many more jobs by 2050.

It is also a renewable, long-lasting energy source, to reiterate. It is caused by the sun's uneven heating, the Earth's diverse

surfaces, and the Coriolis effect! It will stay on Earth for as long as the sun exists and continues to heat the atmosphere. And it can provide energy without polluting the air or utilizing combustion.

Furthermore, an estimated \$2 billion in land-lease payments and state and local taxes are generated annually by wind projects. Communities that produce wind energy can use the additional money to fund local infrastructure projects, lower homeowner taxes, and fund school budgets.

Wind turbines are also cost-effective— some of the cheapest energy sources available, with its cost-competitiveness continually improving.

Finally, they operate in many settings— wherever there is wind, one can utilize wind energy, making it versatile and easy-to-utilize.

As for disadvantages— firstly, wind must compete with other cost-effective energy resources. Although wind and solar are cheaper than most other power plants, wind projects may not always be profitable, especially due to weather conditions.

Again due to weather conditions, however, wind power can be intermittent and, due to several factors, is often only feasible as a true power source in certain areas. Furthermore, for non-distributed wind farms, it may be hard to deliver power to urban areas where it is required, meaning that it may not be feasible everywhere.

Despite this, wind energy stands to be a powerful contender among renewable energy sources, providing clean, versatile energy to many places. With further innovation, integration, and improvement, wind energy

can definitely be a wide-scale and widely used renewable energy.

Geothermal

Geothermal energy, as the name likely suggests, is gathered from the Earth— geo, or Earth, and thermal, related to heat.

All energy stored as heat in the Earth's crust is known as geothermal energy. It is a consistent, low-carbon, renewable heat source that is independent of the weather, primarily used for heating buildings and buildings and generating electricity.

Heat from the earth's crust warms water that is seeped into underground reservoirs. In some places, when water gets hot enough, it will break into the earth's surface as steam or hot water, this is caused by the earth's plates converging and moving.

There are two main ways to extract geothermal energy— geothermal power plants, which utilize heat deep inside the

Earth to generate steam and create electricity, and geothermal heat pumps, which utilize heat near the Earth's surface to heat buildings or water.

Geothermal power plants have three main types, as follows:

Dry Steam power plants: The most commonly used geothermal technology, dry steam power plants use hydrothermal fluids already present in the ground to rotate a turbine which produces electricity. After the steam condenses, it is reinjected into the reservoir.

Flash Steam power plants: In these power plants, hot fluid is pumped into a surface-level tank. As the water cools, it turns into vapor— this is called ‘flashing’— and drives a turbine to power a generator.

Binary Cycle power plants: This method works similarly to the flash steam power plants but with extra steps. Hot fluid from underground heats a secondary fluid known as a heat transfer fluid within a large heat exchanger. This secondary fluid possesses a significantly lower boiling point than the primary fluid and thus vaporizes at a lower temperature. When the second fluid flashes, it powers a turbine to produce electricity.

As technology improves, however, Enhanced Geothermal Systems have been developed— creating artificial reservoirs in hot, dry rock formations. The process drills wells into hot crystalline rock, followed by hydraulically stimulating it to create fractures, then circulating water to absorb heat for power generation. EGS could theoretically unlock geothermal resources

outside of ideal locations— however, this still requires further research and development for public use.

Geothermal Heat Pumps: Much of the heating and cooling of houses and buildings with geothermal energy come from underground with something called a Geothermal Heat Pump (GHP).

Just below the frostline about 3.48 meters down, the earth contains a consistent temperature of around 12 degrees Celsius. We can tap into this energy to provide heating for the winter and cooling for the summer. Bury a loop of pipes called a ‘Heat Exchanger’ just right below the surface. Pour water or water in any free solution into them.

During the winter, the air is frequently colder than the air underground. The solution flows into loops underground and gathers the earth’s heat, this heat is taken to the air and

transmitted into a heat pump, the heat pumps heat the air and then a standard heating system warms the air some more to a comfortable temperature. Finally, ducts cycle the air around the room.

For summer, the system works in reverse. The air on the surface gets considerably warmer than the air underneath, so the fluid in the loops absorbs heat from the building and delivers it down. The ground's lower temperature cools it and cycles it repeatedly.

Geothermal energy, however, has its advantages and disadvantages. Drilling deep into the earth's crust and frostline and installing the necessary equipment requires many funds, taking around \$3000 to \$7000 to install a geothermal power plant per kilowatt, making for a pricey \$3 million-\$7 million to install a power plant— per megawatt.

Even for smaller systems like heat pumps for houses, the cost is exorbitantly high, with the average cost ranging from \$12,000 to \$45,000, depending on the size of the house and energy required.

Furthermore, hot water and steam can only occur in specific places, making the installation of geothermal energy and extraction quite location specific as of now. It works best with places near tectonic plate boundaries, volcanoes, hot springs, et cetera.

Countries with many places that generate steam will greatly benefit from this, but other countries which do not have the same kind of geography may not have enough underground heat. Transporting geothermal energy over long distances is also impractical, making it unsuited for areas not on plate boundaries or with suited places

for extraction. Despite this, technological improvements prove promising to solve this particular hurdle.

Besides this, geothermal reservoirs require careful management to prevent depletion of heat, as excessive extraction of hot water or steam can render the energy source unusable. Sustainable usage involves monitoring and reinjecting fluids to maintain system balance. Additionally, geothermal plants may induce small earthquakes due to pressure changes from deep drilling and water injection, potentially causing minor tremors in geologically active areas, with rare instances of larger seismic events.

Finally, geothermal energy relies on water to generate steam and transfer heat. In areas with limited water supply, this can be a problem. Some plants use closed-loop systems to recycle water, but others may

need large amounts of fresh water, putting pressure on local resources.

However, geothermal energy is also a clean energy source that occurs naturally due to the heat of the Earth. This heat originates from the planet's formation billions of years ago and is continuously replenished by the radioactive decay of minerals in the Earth's crust—exceeding temperatures of 200 degrees Celsius sometimes at depths of only a few kilometres.

Due to its independence of external weather factors, where it is feasible geothermal energy is a reliable power source that can consistently operate, especially in countries with lots of volcanic activity such as Iceland.

Geothermal energy is, for those countries capable of supporting its use,

critical for our shift to renewable energy sources for a greener Earth.

Hydroelectric

Hydroelectricity, the generation of electricity from the energy of moving water, stands as one of humanity's oldest and most significant sources of renewable energy. Its flexibility, reliability, and capacity for large-scale energy storage have made it a cornerstone of modern energy systems, particularly as the world transitions toward low-carbon and sustainable energy sources. However, the development and operation of hydroelectric facilities are not without controversy, as they entail complex trade-offs involving environmental, social, and economic factors.

(Two other chapters are written on branches of hydroelectric energy that could also work well alongside conventional types.)

The harnessing of water's kinetic energy predates the modern era where it began being used for producing electricity. Waterwheels were used as early as Ancient civilisations like Greece, Rome, Egypt, India, and China, utilized for grinding grain, irrigation, and powering simple machines.

During the late 18th and 19th centuries, however, water power became a driving force in the Industrial Revolution. Advanced hydraulic machines were developed, while water frames revolutionized textile manufacturing. Hydraulic power networks developed in the 1840s enabled the transmission of water power to urban centres.

The invention of the electrical generator in the late 19th century then allowed for the coupling of hydraulic turbines with electricity production. The first hydroelectric power

scheme was established in 1878, then another in 1881, and the first Edison hydroelectric station, the Vulcan Street Plant in Appleton, Wisconsin, operating for the first time in 1882 with an output of 12.5 kilowatts.

By 1886, there were 45 hydroelectric stations in the US and Canada combined; three years later the United States alone had 200 facilities. The early 1900s also brought a proliferation of small hydroelectric plants.

The latter half of the 20th century and the early 21st century saw the construction of even larger hydroelectric facilities. For instance, the Three Gorges Dam, completed in China in 2012 spanning the Yangtze River, has a capacity of 22.5 gigawatts.

Today, hydropower remains the largest source of renewable electricity worldwide, with a global installed capacity of nearly 1,400 GW as of 2023.

Hydroelectric power generation is based on the conversion of the potential and kinetic energy of water into mechanical energy, and subsequently into electrical energy.

Water is stored at a higher elevation in a reservoir behind a dam. The difference in height between the water surface in the reservoir and the outflow point (the “head”) determines the potential energy available for conversion. Then, the water is released from the reservoir through a large pipe called a penstock, which directs the flow toward the turbine. The volume of water (flow rate) and the head are the two primary factors influencing power output.

As water flows through a turbine, it spins the blades, converting the water’s kinetic and potential energy into mechanical rotational energy. The rotating turbine shaft is

connected to a generator, which converts mechanical energy into electrical energy via electromagnetic induction, similarly to most or all turbine-facilitated generators. The generated electricity is stepped up in voltage by transformers and transmitted via power lines to homes, businesses, and industries.

The theoretical power output of a hydroelectric plant can be approximated by the formula:

$$P = \eta \rho g Q h$$

where:

P = Power output (watts)

η = Efficiency of the system (typically 0.7–0.95 for modern turbines)

ρ = Density of water ($\sim 1000 \text{ kg/m}^3$)

g = Acceleration due to gravity (9.8 m/s^2)

Q = Volumetric flow rate (m^3/s)

h = Head (meters)

As for the types of hydroelectric plants:

Conventional (Impoundment)

Hydroelectric Plants: The most common type of hydroelectric plants, these involve a dam that impounds a river to create a large reservoir, water being released as needed to generate electricity. They have a high capacity, ability to store water for peak demand, and are multipurpose. However, they have more environmental impact—the building of the Three Gorges Dam contributed to environmental damage along the Yangtze River and environs.

Run-of-the-River (Diversion)

Plants: These facilities channel a portion of river flow to turbines, often with little or no reservoir storage. They have lower environmental impact with minimal land inundation and are suitable for smaller rivers.

However, output varies with river flow and they have a limited ability to store energy.

Pumped-Storage Hydropower: During periods of low electricity demand, excess energy is used to pump water from a lower reservoir to an upper reservoir. During peak demand, water is released back down to generate electricity. It provides grid stability, load balancing and large-scale energy storage. However, they lose energy due to pumping losses, require a lot of capital, and have many site-specific requirements.

Conduit Hydropower: These generate electricity from water flowing through man-made conduits like irrigation canals or municipal water system. They utilise existing infrastructure with minimal environmental impact, but they are typically quite small-scale.

Small, Micro, Pico

Hydropower: Although definitions may vary by country, small hydro are 10-50 megawatts based on country, micro are $\leq 100\text{kW}$, while pico are $\leq 5\text{kW}$. They have a low environmental impact and help with local economic development.

Tidal and other Marine-based Hydropower will be further elaborated on in later chapters.

Despite locational constraints, hydropower is and will likely continue to be one of the greatest contenders in renewable energy fields.

Though a downside to building such a massive dam is the massive building costs, with the Three Gorges Dam requiring a ridiculous 28 million cubic meters of concrete alone.

Another issue is that such massive quantities of water can change the axis of the earth, in fact when the three gorges dam was first fully filled, it changed the angle of earth's tilt slightly.

It is a clean, renewable source of energy that will never run out—it does not consume water, instead harnessing the kinetic energy of its movement.

However, it is generally location-specific— if artificially built, large commercial-scale plants require not only large amounts of energy to create reservoirs but also external water sources to fill them, which may put stress on local water sources. Smaller, less environmentally impactful plants may not be able to provide enough energy to be reliable as a source.

Nuclear

Nuclear energy has the most ambiguous position in renewable energy terms. Some argue that nuclear energy—being facilitated by nuclear elements such as radium or uranium which are non-renewable—is also a non-renewable source of energy. Others state that since nuclear elements can be reused, it counts as a renewable source of energy—or that developing nuclear fusion can almost eliminate these issues.

Then, people argue about whether or not it counts as clean energy—they cite nuclear accidents like the Chernobyl or Fukushima disasters which released radioactive waste; while others rebut that it was caused by various failures in mechanisms or human error.

After that, they quote safety— alpha particles in the body wreaking havoc with anaemia and leukaemia— but alpha particles cannot penetrate skin, and proper safety measures mean that there is little to no true safety risk.

Nuclear power works through either nuclear fission or nuclear fusion. As of right now, nuclear fission is generally used, while nuclear fusion holds little true application. China's Academy of Sciences recently announced that its 'artificial sun' nuclear fusion reactor recently overcame another major obstacle, however, proving promising.

Nuclear Fission: This is a reaction where the nucleus of an atom splits into two or more smaller nuclei, releasing energy.

For example, the nucleus of a uranium-235 atom, when hit by a neutron, will split into two smaller nuclei as well as some

excess neutrons, which will then hit other surrounding uranium-235 atoms and so on and so forth, causing a chain reaction. When this occurs, energy is released in the form of heat and radiation, and the heat can be converted into electricity. After being used, fuel must be safely disposed of, but it can later be reused.

This can also be used with many other elements with uranium being only one example. Another element that can be used is plutonium — when uranium reaches its half-life, it degrades into neptunium which quickly decays into plutonium.

Nuclear Fusion: In nuclear fusion, nuclear reactions between lighter elements form heavier elements. (Nuclear fusion is also the same reaction that powers the sun.) The total mass is less than the two that formed it— the ‘missing’ given off as energy

in the equation $E=mc^2$. Nuclear fusion has much less risk and much more energy per kilogram of fuel. So far, the most promising combination for power on Earth today is the fusion of a deuterium and tritium atom. It requires temperatures of around 39 million degrees Celsius, producing 17.6 million electron volts of energy.

Nuclear energy provides over 20% of the world's low carbon electricity, with little to no carbon emissions during the process of nuclear fission. However, the radioactive waste can be dangerous if not properly disposed of— another reason why the development of nuclear fusion technology is so important. Nuclear fusion does not produce radioactive waste as it depends on high amounts of pressure and heat to convert neutrons into energy directly.

Here are some of the types of nuclear reactors:

Pressurized Water Reactors: The reactor core generates heat through nuclear reactions. Then, water in the primary circuit absorbs thermal energy, obstructed from boiling by the high pressure. The heated water from the primary circuit flows to a steam generator to power a turbine. After this, the steam is condensed and flows back to the beginning.

Boiling Water Reactors: In boiling water reactors, water is heated directly inside of the reactor core until boiling— so the steam comes directly from inside the reactor. This eliminates the need for a secondary circuit and simplifies heat transfer.

Gas-Cooled Nuclear Reactor: In gas-cooled reactors, an inert gas— typically

helium or carbon dioxide— is used as the primary coolant to remove heat produced by nuclear reactions in the core. They also often use graphite moderators to slow down neutrons for more efficient fission and to maintain the chain reaction. Later, the heat in the gas can be converted into steam to power turbines.

Especially with the development of nuclear fusion, nuclear power is a strong contender among renewable or clean energy sources. Nuclear power also actually rarely has problems or issues— but when it does, they are on the scale of those in Fukushima or Chernobyl. Huge.

Due to the amount of energy produced by causing changes on an atomic or subatomic level, nuclear power plants not only require significantly less land area than other renewable energy sources like wind or

solar, but also much less fuel to generate the same energy as several tons of coal. Nuclear power plants also operate continuously, able to produce energy regardless of weather conditions.

However, the large scale of even the few nuclear accidents that occur as well as their lasting impact raises significant concerns. The radioactive waste generated by nuclear fission also requires careful management and long-term storage, while nuclear power plants require substantial upfront investments.

Furthermore, while fuel for nuclear fission is more abundant than fossil fuels, they are still non-renewable, and advancements in recycling technology are necessary.

Finally, due to association with nuclear weapons and past accidents, public

perception of nuclear power is overwhelmingly negative.

Although much of our clean energy even now comes from nuclear sources, public perception plays a key role in limiting its progress.

Despite this, especially with the development of nuclear fusion technologies, nuclear energy has proven to be a promising energy source.

Hydrogen

Hydrogen power refers to energy generated from hydrogen, either by combustion or through fuel cells, where hydrogen reacts with oxygen to produce electricity and water without carbon emissions.

Although hydrogen is an abundant element, on Earth it must be extracted from compounds like water through processes such as electrolysis, which utilizes electricity and a conductor to extract certain elements like hydrogen from water, requiring energy. This makes hydrogen have less net energy production than other energy sources, necessitating a lot of electricity.

Hydrogen is also extremely volatile and combustible, as well as being a gas at room

temperature. This makes it difficult to properly transport.

Hydrogen power, similar to liquefied natural gas outputs energy through combustion like in combustion engines.

A hydrogen combustion engine is a modified internal combustion engine which burns hydrogen as opposed to a standard internal combustion engine which burns gasoline.

Like a normal internal combustion engine, hydrogen combustion engines rely on pistons and controlled explosions to turn an axle.

Hydrogen combustion engines have many advantages over traditional internal combustion engines, including the fact that hydrogen emits no carbon gases and only releases water vapour when burned.

Hydrogen's volatile reactions allow a lesser amount to be used for the same power output in a hydrogen internal combustion engine.

The main issue with hydrogen internal combustion engines is that they are difficult to engineer, in the past hydrogen powered vehicles such as the De Rivaz engine from 1806 is that they require extreme modification in order to handle hydrogen's immense power, as well as how hydrogen's low density lowering maximum tank capacity.

The electrolysis of water to decompose it into hydrogen and oxygen requires an external energy source, typically direct current electricity. An electric current is passed through water containing an electrolyte, facilitating the reaction.

The chemical $\text{H}_2\text{O} \rightarrow \text{H}_2 + \text{O}_2$ can be balanced into $2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{H}_2(\text{g}) + \text{O}_2(\text{g})$.

Hydrogen is also coming back into prevalence as a fuel source— completely clean, it produces only water and heat, reducing greenhouse gas emissions and improving air quality.

There is also potential for splitting the hydrogen and carbon in hydrocarbons (the main compounds found in fossil fuels) and utilizing hydrogen for energy whilst the carbon can be used for many other things like making graphene.

This exceedingly simple chemical formula H_2O becomes the basis for a powerful energy source.

Hydrogen energy, though volatile, is simple and efficient, one of the cleanest energy sources in the world.

Rain

Singaporean researchers recently created systems that capture the kinetic energy of raindrops and are quite literally 10 times more efficient than conventional hydroelectric power. Though controlled by weather, all rain or water falling can generate energy—whether downpours of flood scales or light morning mists.

The first way this works is with something called ‘plug flow’. These systems can convert over 10% of the energy from falling water—it sounds small, but it is five orders of magnitude (100,000 times) more output than methods using continuous water flow for charge separation. The system utilizes small, vertical tubes on millimetre scales, and as water falls through, short air pockets are formed, interspersed with the water. As the water comes into contact with

the inner surface of the tube, charge separation occurs— negatively charged hydroxide ions tend to adhere to the tube's surface, whilst positively charged hydrogen ions follow the water. The separation of these charges generates an electric current.

The other method utilizes TENGs, or Triboelectric Nanogenerators. When a raindrop lands on the hydrophobic, nanostructured surface, it spreads briefly and slides off. This generates charges through the triboelectric effect— electric charge is transferred between two objects when they contact or slide against each other, producing static electricity.

This method, many times more efficient with far less environmental impact than conventional hydropower systems, proves as a testament to the creativity and genius of Singapore's scientists and engineers.

Piezoelectric

Japan recently harnessed power from footsteps with piezoelectric technology.

Piezoelectricity is a process in which certain materials generate electric charge when mechanical stress is applied. This phenomenon occurs due to the displacement of positive and negative charge centres within the material's structure leading to electric polarization and the generation of voltage.

Discovered in 1880 by a pair of French physicists during a series of experiments with quartz crystal, this technology holds much potential for use in renewable energy fields—and that potential has begun being uncovered by the piezoelectric tiles in Japan.

The piezoelectric effect has been witnessed in several materials such as

quartz, Rochelle salt, and certain ceramics. This is due to their crystalline structure allowing electric charges within the lattice to shift when stressed, creating an electric dipole. If an electric field is applied, the lattice will also deform.

Although these tiles generate only small amounts of electricity, placing them in areas with large amounts of foot traffic allows them to be used as a good supplementary energy source.

Lightning

Lightning energy is a theoretically possible but practically non-viable renewable source of energy.

People have been investigating the possibility of harnessing lightning for electricity for decades, since a single bolt of lightning carries a huge amount of energy (over 5 billion joules or 5 gigajoules of energy, which can provide a household with all their electrical needs for a month). And a single thunderstorm can produce the same amount of energy found in an atom bomb.

Despite the immense amount of energy generated contributing to its appeal and potential, there are many drawbacks to this. Such as the limitations of our current technology, safety considerations and even the bolt(s) of lightning itself have been

massive hurdles for the practical applications of this endeavour.

The extreme electrical power released by one single lightning bolt is only concentrated on that one small location that it struck, and it is passed in an extremely short period of time (in the span of microseconds). Combined that with the huge amount of electrical power that it carries, that means we must develop technology that is not only able to withstand high current intensities but also store and utilize that power.

However, lightning is in and of itself electricity. It exists as a charge— that's why it causes electrocution. The reason why it produces heat is due to resistance.

This is why our team decided to construct our model based on lightning

energy. There is little necessity for conversion through turbines, as well.

Outside of this, a majority of power outages happen during thunderstorms— lightning energy could be a supplementary source of power during outages.

The model, though shown, is not all— we operate on the principle that lightning is a current. The lightning is conducted through thick rods. As it travels, any heat it produces is absorbed in tanks with sand and water, equipped with turbines in case of steam. Further down, resistors are also used to help curb the voltage. Finally, the reduced energy can be safely sent for usage.

In the model, we used a power cell to represent lightning, an LED strip to represent usage of the energy, and did not truly make

tanks. This is because we were unable to find the materials to create a generator, and therefore the tank in our model does not serve a purpose in the circuitry. Also, we were unable to generate a large enough amount of charge to produce heat energy.

We understand that this model may not be scalable or feasible, but it is our theoretical model constructed with our knowledge of electricity and circuitry.

If lightning can be properly harnessed, it will be groundbreaking in the field of renewable energy.